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library(tidyverse)
library(httr)
library(jsonlite)
library(bea.R)
library(readxl)
library(curl)

BLS_KEY    <- Sys.getenv("BLS_API_KEY")
BEA_KEY    <- Sys.getenv("BEA_API_KEY")
CENSUS_KEY <- Sys.getenv("CENSUS_API_KEY")

# Training years: 2013–2016 | Test year: 2024
# 2017–2023 are intentionally excluded.
train_years <- 2013:2016
test_year   <- 2024
all_years   <- c(train_years, test_year)

# =====
# FUNCTIONS TO PULL DATA
# =====

## BLS API -----
pull_bls <- function(series_ids, start_year, end_year) {
  body <- list(
    seriesid      = series_ids,
    startyear     = as.character(start_year),
    endyear       = as.character(end_year),
    registrationkey = BLS_KEY,
    annualaverage = TRUE
  )
  res <- POST("https://api.bls.gov/publicAPI/v2/timeseries/data/",
    body = toJSON(body, auto_unbox = TRUE),
    content_type_json())
  content(res, as = "parsed")
}

### LAUS -----
laus_series <- c(
  "LAUST12000000000000009", # CNP
  "LAUST12000000000000006", # LFPR
  "LAUST12000000000000008"  # Total Labor Force
)

# BLS API only accepts a contiguous range; pull training window and
# test year separately then combine.
laus_train <- pull_bls(laus_series, min(train_years), max(train_years))
laus_test  <- pull_bls(laus_series, test_year, test_year)

### QCEW -----
pull_qcew <- function(year) {
  url <- paste0(
    "https://data.bls.gov/cew/data/api/",
    year, "/a/area/12000.csv"
  )
  tryCatch({
    read_csv(url, show_col_types = FALSE) %>% mutate(year = year)
  }, error = function(e) {
    if (year == 2013)
      message("Note: QCEW 2013 not available via BLS API (404) – ",
        "qcew features for 2013 will be imputed by step_impute_median()")
    else
      message("Skipping QCEW year ", year, ": ", e$message)
    NULL
  })
}

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})
}

qcew_raw <- map_dfr(all_years, pull_qcew)

### CES -----
ces_series <- c(
  "CEU1000000011", # Natural Resources & Mining
  "CEU2000000011", # Construction
  "CEU3000000011", # Manufacturing
  "CEU4000000011", # Trade, Transportation & Utilities
  "CEU5000000011", # Information
  "CEU5500000011", # Financial Activities
  "CEU6000000011", # Professional & Business Services
  "CEU6500000011", # Education & Health Services
  "CEU7000000011", # Leisure & Hospitality
  "CEU8000000011" # Other Services
)

ces_train <- pull_bls(ces_series, min(train_years), max(train_years))
ces_test  <- pull_bls(ces_series, test_year, test_year)

# =====
# OEWS RESEARCH ESTIMATES – TARGET VARIABLE (manually downloaded)
# 2013:      Data 2/2013_sectors/oes_research_2013_sec_*.xlsx (multiple files)
# 2014–2016: Data 2/oes_research_{year}_allsectors.xlsx
# 2024:      Data 2/oes_research_2024_allsectors.xlsx
# =====

# Read years stored as one file per NAICS sector group (2013, 2014, 2015)
read_ows_sector_year <- function(year) {
  dir <- paste0("Data 2/", year, "_sectors")
  sector_files <- list.files(dir, pattern = "\\\\.xlsx$", full.names = TRUE)
  if (length(sector_files) == 0) {
    message("No sector files found in ", dir, " – skipping ", year)
    return(NULL)
  }
  message("Reading OEWS research estimates for ", year,
    " (", length(sector_files), " sector files)...")
  map_dfr(sector_files, function(f) {
    read_xlsx(f, guess_max = 10000) %>%
    mutate(across(everything(), as.character))
  }) %>%
  mutate(year = as.integer(year))
}

# Read a single allsectors file (2016, 2024)
read_ows_file <- function(year) {
  path <- paste0("Data 2/oes_research_", year, "_allsectors.xlsx")
  if (!file.exists(path)) {
    message("Missing: ", path)
    return(NULL)
  }
  message("Reading OEWS research estimates for ", year, "...")
  read_xlsx(path, guess_max = 10000) %>%
  mutate(across(everything(), as.character)) %>%
  mutate(year = as.integer(year))
}

ows_research_raw <- bind_rows(
  map_dfr(c(2013L, 2014L, 2015L), read_ows_sector_year), # sector files
  map_dfr(c(2016L, test_year), read_ows_file)          # allsectors files
)

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message("---- OEWS RAW diagnostics ----")
message("Years in row: ", paste(sort(unique(oews_research_raw$year)), collapse = ", "))
message("Rows per year:")
print(table(oews_research_raw$year))
message("Columns present: ", paste(names(oews_research_raw), collapse = ", "))

for (yr in sort(unique(oews_research_raw$year))) {
  sub      <- oews_research_raw[oews_research_raw$year == yr, ]
  at_col   <- if ("area_title" %in% names(sub)) sub[["area_title"]] else
              if ("AREA_TITLE" %in% names(sub)) sub[["AREA_TITLE"]] else NA_character_
  grp_col  <- if ("o_group" %in% names(sub)) sub[["o_group"]] else
              if ("O_GROUP" %in% names(sub)) sub[["O_GROUP"]] else
              if ("group" %in% names(sub)) sub[["group"]] else NA_character_
  at_vals  <- paste(head(unique(na.omit(at_col)), 3), collapse = " / ")
  grp_vals <- paste(head(unique(na.omit(grp_col)), 5), collapse = " / ")
  message(sprintf(" %d: %d rows | area_title: [%s] | group col: [%s]",
                  yr, nrow(sub), at_vals, grp_vals))
}
message("---- end OEWS RAW diagnostics ----")

# Helper: ensure all expected column variants exist in df before coalescing.
# Adds missing columns as NA_character_ using direct assignment (safe for any df size).
pad_oews_cols <- function(df) {
  expected <- c("AREA_TITLE", "area_title",
                "O_GROUP", "o_group", "group",
                "NAICS", "naics",
                "NAICS_TITLE", "naics_title",
                "OCC_CODE", "occ code",
                "OCC_TITLE", "occ title",
                "TOT_EMP", "tot_emp",
                "EMP_PRSE", "emp_prse",
                "H_MEAN", "h_mean",
                "A_MEAN", "a_mean",
                "I_GROUP", "i_group")
  for (col in setdiff(expected, names(df))) {
    df[[col]] <- NA_character_
  }
  df
}

oews_padded <- pad_oews_cols(oews_research_raw)

message("---- post-pad column check ----")
message("o_group present: ", "o_group" %in% names(oews_padded))
message("O_GROUP present: ", "O_GROUP" %in% names(oews_padded))
message("group present: ", "group" %in% names(oews_padded))
message("Sample group values by year:")
for (yr in sort(unique(oews_padded$year))) {
  g <- unique(na.omit(oews_padded$group[oews_padded$year == yr]))
  o <- unique(na.omit(oews_padded$O_GROUP[oews_padded$year == yr]))
  message(sprintf(" %d: group=[%s] O_GROUP=[%s]", yr,
                  paste(head(g, 4), collapse="/"), paste(head(o, 4), collapse="/")))
}

oews_research_clean <- oews_padded %>%
  mutate(
    area_title_final = tolower(trimws(coalesce(AREA_TITLE, area_title))),
    o_group_final    = coalesce(
      tolower(trimws(O_GROUP)),
      tolower(trimws(o_group)),
      tolower(trimws(group)),
      if_else(year %in% 2013:2015, "detailed", NA_character_) # sector files only contain
detailed rows
    ),
    naics_final      = coalesce(NAICS, naics),

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    naics_title_final = coalesce(NAICS_TITLE, naics_title),
    occ_code_final    = coalesce(OCC_CODE, `occ code`),
    occ_title_final   = coalesce(OCC_TITLE, `occ title`),
    tot_emp_final     = coalesce(TOT_EMP, tot_emp),
    emp_prse_final    = coalesce(EMP_PRSE, emp_prse),
    h_mean_final      = coalesce(H_MEAN, h_mean),
    a_mean_final      = coalesce(A_MEAN, a_mean),
    i_group_final     = coalesce(I_GROUP, i_group)
  )

message("---- after mutate, before filter ----")
message("o_group_final unique values: ",
paste(head(unique(na.omit(oews_research_clean$o_group_final)), 8), collapse=" / "))
message("Rows per year (pre-filter):")
print(table(oews_research_clean$year))

oews_research_clean <- oews_research_clean %>%
  filter(area_title_final == "florida",
         o_group_final    %in% c("detailed", "detail")) %>%
  select(Year          = year,
         naics          = naics_final,
         naics_title    = naics_title_final,
         i_group        = i_group_final,
         occ_code       = occ_code_final,
         occ_title      = occ_title_final,
         tot_emp        = tot_emp_final,
         emp_prse       = emp_prse_final,
         h_mean         = h_mean_final,
         a_mean         = a_mean_final) %>%
  mutate(across(c(tot_emp, emp_prse, h_mean, a_mean),
                 ~suppressWarnings(as.numeric(.x))))

message("OEWS years present: ", paste(sort(unique(oews_research_clean$Year)), collapse =
", "))

## BEA API -----
# Cache BEA results to disk – the API rate-limits at ~429 after many calls.
# On first run this fetches everything and saves to data/bea_cache.rds.
# On all subsequent runs it loads instantly from cache.
# Delete data/bea_cache.rds only if you need to refresh the BEA data.

bea_cache_path <- "data/bea_cache.rds"

if (file.exists(bea_cache_path)) {

  message("Loading BEA data from cache: ", bea_cache_path)
  bea_cache <- readRDS(bea_cache_path)
  bea_nominal_gsp <- bea_cache$bea_nominal_gsp
  bea_nominal_gdp_industry <- bea_cache$bea_nominal_gdp_industry
  bea_real_gsp <- bea_cache$bea_real_gsp
  bea_real_gdp_industry <- bea_cache$bea_real_gdp_industry
  bea_pce <- bea_cache$bea_pce
  bea_gov_gdp <- bea_cache$bea_gov_gdp
  bea_construction_gdp <- bea_cache$bea_construction_gdp
  bea_employment <- bea_cache$bea_employment
  bea_deflator <- bea_cache$bea_deflator
  bea_line_codes <- bea_cache$bea_line_codes

} else {

  message("Fetching BEA data from API (first run – will cache to ", bea_cache_path, ")")

  bea_line_codes <- beaParamVals(BEA_KEY, "Regional", "LineCode") %>%
    as.data.frame() %>%

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filter(grepl("SAGDP2", ParamValue.Desc)) %>%
filter(endsWith(ParamValue.Desc, "")) %>%
mutate(ParamValue.Key = as.numeric(ParamValue.Key))

pull_bea <- function(table_name, line_code, years = all_years) {
  params <- list(
    UserID      = BEA_KEY,
    Method      = "GetData",
    DatasetName = "Regional",
    TableName   = table_name,
    LineCode    = line_code,
    GeoFIPS     = "STATE",
    Year        = paste(years, collapse = ","),
    ResultFormat = "json"
  )
  beaGet(params, asWide = FALSE) %>%
  filter(GeoName == "Florida")
}

# Nominal GSP - Total
bea_nominal_gsp <- pull_bea("SAGDP2", line_code = 1) %>%
  mutate(variable = "nominal_gsp") %>%
  select(TimePeriod, DataValue, variable)

# Nominal GDP by Industry
bea_nominal_gdp_industry <- map_dfr(3:86, function(lc) {
  Sys.sleep(1) # 1s between calls to stay under BLS rate limit
  tryCatch({
    pull_bea("SAGDP2", line_code = lc) %>% mutate(line_code = lc)
  }, error = function(e) {
    message("Skipping SAGDP2 line code ", lc, ": ", conditionMessage(e))
    NULL
  })
}) %>%
  left_join(bea_line_codes, by = join_by(line_code == ParamValue.Key)) %>%
  mutate(ParamValue.Desc = paste0(sub(".*: ", "", ParamValue.Desc))) %>%
  select(TimePeriod, DataValue, ParamValue.Desc, line_code) %>%
  rename(Year = TimePeriod) %>%
  mutate(Year = as.integer(Year))

# Real GSP - Total
bea_real_gsp <- pull_bea("SAGDP9", line_code = 1) %>%
  mutate(variable = "real_gsp") %>%
  select(TimePeriod, DataValue, variable)

# Real GDP by Industry
bea_real_gdp_industry <- map_dfr(3:86, function(lc) {
  Sys.sleep(1)
  tryCatch({
    pull_bea("SAGDP9", line_code = lc) %>% mutate(line_code = lc)
  }, error = function(e) {
    message("Skipping SAGDP9 line code ", lc, ": ", conditionMessage(e))
    NULL
  })
}) %>%
  left_join(bea_line_codes, by = join_by(line_code == ParamValue.Key)) %>%
  mutate(ParamValue.Desc = paste0(sub(".*: ", "", ParamValue.Desc))) %>%
  select(TimePeriod, DataValue, ParamValue.Desc, line_code) %>%
  rename(Year = TimePeriod) %>%
  mutate(Year = as.integer(Year))

# Personal Consumption Expenditures - Total
bea_pce <- pull_bea("SAPCE1", line_code = 1) %>%
  mutate(variable = "pce") %>%
  select(TimePeriod, DataValue, variable)

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# Government GDP - line code 83
bea_gov_gdp <- pull_bea("SAGDP2", line_code = 83) %>%
  mutate(variable = "government_gdp") %>%
  select(TimePeriod, DataValue, variable)

# Construction GDP - line code 8
bea_construction_gdp <- pull_bea("SAGDP2", line_code = 8) %>%
  mutate(variable = "construction_gdp") %>%
  select(TimePeriod, DataValue, variable)

# BEA Employment
bea_employment <- pull_bea("SAINC4", line_code = 7010) %>%
  mutate(variable = "total_employment") %>%
  select(TimePeriod, DataValue, variable)

# National Price Deflator
pull_nipa <- function(table_name, years = all_years) {
  params <- list(
    UserID      = BEA_KEY,
    Method      = "GetData",
    DatasetName = "NIPA",
    TableName   = table_name,
    Frequency   = "A",
    Year        = paste(years, collapse = ","),
    ResultFormat = "json"
  )
  beaGet(params, asWide = FALSE)
}

bea_deflator <- pull_nipa("T10109") %>%
  filter(LineNumber == "1") %>%
  select(TimePeriod, DataValue) %>%
  rename(year = TimePeriod, gdp_deflator = DataValue) %>%
  mutate(year = as.integer(year))

# Save to cache
saveRDS(
  list(
    bea_nominal_gsp      = bea_nominal_gsp,
    bea_nominal_gdp_industry = bea_nominal_gdp_industry,
    bea_real_gsp         = bea_real_gsp,
    bea_real_gdp_industry = bea_real_gdp_industry,
    bea_pce              = bea_pce,
    bea_gov_gdp          = bea_gov_gdp,
    bea_construction_gdp = bea_construction_gdp,
    bea_employment       = bea_employment,
    bea_deflator         = bea_deflator,
    bea_line_codes       = bea_line_codes
  ),
  bea_cache_path
)
message("BEA data cached to ", bea_cache_path)
}

# =====
# CLEAN DATA
# =====

## LAUS -----
# Combine training and test year pulls, then parse
parse_bls_response <- function(raw) {
  map_dfr(raw$Results$series, function(series) {

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    map_dfr(series$data, function(obs) {
      tibble(
        series_id = series$seriesID,
        year      = as.integer(obs$year),
        period    = obs$period,
        value     = as.numeric(obs$value)
      )
    })
  })
}

laus_clean <- bind_rows(
  parse_bls_response(laus_train),
  parse_bls_response(laus_test)
) %>%
  filter(period == "M13") %>%
  select(-period) %>%
  mutate(variable = case_when(
    series_id == "LAUST1200000000000009" ~ "cnp",
    series_id == "LAUST1200000000000006" ~ "lfpr",
    series_id == "LAUST1200000000000008" ~ "labor_force"
  )) %>%
  select(-series_id) %>%
  pivot_wider(names_from = variable, values_from = value) %>%
  rename(Year = year)

## QCEW -----
assign_ces_supersector <- function(naics_code) {
  naics_2digit <- as.integer(substr(as.character(naics_code), 1, 2))
  case_when(
    naics_2digit %in% c(11, 21) ~ "CEU1000000",
    naics_2digit == 23 ~ "CEU2000000",
    naics_2digit %in% 31:33 ~ "CEU3000000",
    naics_2digit %in% c(22, 42, 44, 45, 48, 49) ~ "CEU4000000",
    naics_2digit == 51 ~ "CEU5000000",
    naics_2digit %in% 52:53 ~ "CEU5500000",
    naics_2digit %in% 54:56 ~ "CEU6000000",
    naics_2digit %in% 61:62 ~ "CEU6500000",
    naics_2digit %in% 71:72 ~ "CEU7000000",
    naics_2digit == 81 ~ "CEU8000000",
    naics_2digit %in% 91:93 ~ NA_character_,
    TRUE ~ NA_character_
  )
}

qcew_clean <- qcew_raw %>%
  filter(own_code == 5, agglvl_code == 56) %>%
  select(year, industry_code, annual_avg_emplvl, avg_annual_pay) %>%
  filter(avg_annual_pay != 0) %>%
  mutate(ces_series_id = assign_ces_supersector(industry_code)) %>%
  rename(Year = year)

## CES -----
ces_clean <- bind_rows(
  parse_bls_response(ces_train),
  parse_bls_response(ces_test)
) %>%
  filter(period == "M13") %>%
  mutate(year = as.integer(year), value = as.numeric(value)) %>%
  select(year, seriesID = series_id, avg_weekly_hours = value) %>%
  rename(Year = year) %>%
  mutate(seriesID = substr(seriesID, 1, 10))

# =====

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# MERGE DATA SETS
# =====

bea_cleaner <- function(df) {
  df %>%
    rename(Year = TimePeriod,
            !!df$variable[1] := DataValue) %>%
    mutate(Year = as.integer(Year)) %>%
    select(-variable)
}

bea_construction_gdp_clean <- bea_cleaner(bea_construction_gdp)
bea_employment_clean <- bea_cleaner(bea_employment)
bea_gov_gdp_clean <- bea_cleaner(bea_gov_gdp)
bea_nominal_gsp_clean <- bea_cleaner(bea_nominal_gsp)
bea_pce_clean <- bea_cleaner(bea_pce)
bea_real_gsp_clean <- bea_cleaner(bea_real_gsp)

bea_deflator_clean <- bea_deflator %>% rename(Year = year)
bea_nominal_gdp_industry_clean <- bea_nominal_gdp_industry %>%
  rename(nominal_gdp_industry = DataValue)
bea_real_gdp_industry_clean <- bea_real_gdp_industry %>%
  rename(real_gdp_industry = DataValue)

bea_gdp_industry <- left_join(
  bea_real_gdp_industry_clean,
  bea_nominal_gdp_industry_clean,
  by = join_by(Year, ParamValue.Desc, line_code)
)

bea_clean <- list(
  bea_construction_gdp_clean,
  bea_employment_clean,
  bea_gov_gdp_clean,
  bea_nominal_gsp_clean,
  bea_pce_clean,
  bea_real_gsp_clean,
  bea_deflator_clean,
  bea_gdp_industry
) %>%
  reduce(full_join, by = "Year")

# =====
# CROSSWALK: NAICS 2-Digit <-> CES Supersector <-> BEA Line Code
# =====
naics_ces_bea_crosswalk <- tribble(
  ~naics_2digit, ~naics_sector_title,
  ~ces_series_id, ~ces_sector_title, ~bea_line_code,
  "11", "Agriculture, Forestry, Fishing and Hunting",
  "CEU1000000", "Natural Resources & Mining", 3,
  "21", "Mining, Quarrying, and Oil and Gas Extraction",
  "CEU1000000", "Natural Resources & Mining", 6,
  "22", "Utilities",
  "CEU4000000", "Trade, Transportation & Utilities", 7,
  "23", "Construction",
  "CEU2000000", "Construction", 8,
  "31", "Manufacturing",
  "CEU3000000", "Manufacturing", 9,
  "32", "Manufacturing",
  "CEU3000000", "Manufacturing", 9,
  "33", "Manufacturing",
  "CEU3000000", "Manufacturing", 9,
  "42", "Wholesale Trade",
  "CEU4000000", "Trade, Transportation & Utilities", 12,

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"44", "Retail Trade",	
"CEU4000000", "Trade, Transportation & Utilities",	13,
"45", "Retail Trade",	
"CEU4000000", "Trade, Transportation & Utilities",	13,
"48", "Transportation and Warehousing",	
"CEU4000000", "Trade, Transportation & Utilities",	14,
"49", "Transportation and Warehousing",	
"CEU4000000", "Trade, Transportation & Utilities",	14,
"51", "Information",	
"CEU5000000", "Information",	23,
"52", "Finance and Insurance",	
"CEU5500000", "Financial Activities",	25,
"53", "Real Estate and Rental and Leasing",	
"CEU5500000", "Financial Activities",	30,
"54", "Professional, Scientific, and Technical Services",	
"CEU6000000", "Professional & Business Services",	34,
"55", "Management of Companies and Enterprises",	
"CEU6000000", "Professional & Business Services",	35,
"56", "Administrative and Waste Management Services",	
"CEU6000000", "Professional & Business Services",	36,
"61", "Educational Services",	
"CEU6500000", "Education & Health Services",	39,
"62", "Health Care and Social Assistance",	
"CEU6500000", "Education & Health Services",	40,
"71", "Arts, Entertainment, and Recreation",	
"CEU7000000", "Leisure & Hospitality",	46,
"72", "Accommodation and Food Services",	
"CEU7000000", "Leisure & Hospitality",	47,
"81", "Other Services (except Public Administration)",	
"CEU8000000", "Other Services",	50,
"92", "Public Administration",	
NA_character_, "Government (not in CES private)",	51

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# =====
# STEP 1: Aggregate QCEW to NAICS 2-Digit Sector Level
# =====
qcew_by_naics <- qcew_clean %>%
  mutate(naics_2digit = substr(as.character(industry_code), 1, 2)) %>%
  filter(!is.na(naics_2digit)) %>%
  group_by(Year, naics_2digit) %>%
  summarise(
    qcew_total_emp = sum(annual_avg_emplvl, na.rm = TRUE),
    qcew_wtd_avg_pay = weighted.mean(avg_annual_pay, w = annual_avg_emplvl, na.rm = TRUE),
    .groups = "drop"
  )

# =====
# STEP 2: Map BEA Industry GDP to NAICS 2-Digit Sectors
# =====
bea_by_naics <- bea_gdp_industry %>%
  inner_join(
    naics_ces_bea_crosswalk %>% select(naics_2digit, bea_line_code) %>% distinct(),
    by = c("line_code" = "bea_line_code")
  ) %>%
  select(Year, naics_2digit, real_gdp_industry, nominal_gdp_industry)

# =====
# STEP 3: Attach Crosswalk Keys to OEWS
# =====
oews_with_keys <- oews_research_clean %>%
  mutate(
    naics_2digit = case_when(
      is.na(naics) | naics == "000000" ~ NA_character_,
      TRUE ~ substr(naics, 1, 2)
    )
  )

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    )
  ) %>%
  left_join(
    naics_ces_bea_crosswalk %>%
      select(naics_2digit, ces_series_id, naics_sector_title),
    by = "naics_2digit"
  )

# =====
# STEP 4: Aggregate OEWS to Year x NAICS 2-Digit Sector x Occupation
# =====
oews_by_naics <- oews_with_keys %>%
  filter(!is.na(naics_2digit)) %>%
  group_by(Year, naics_2digit, naics_sector_title, occ_code, occ_title) %>%
  summarise(
    oews_tot_emp = sum(tot_emp, na.rm = TRUE),
    oews_h_mean = weighted.mean(h_mean, w = tot_emp, na.rm = TRUE),
    oews_a_mean = weighted.mean(a_mean, w = tot_emp, na.rm = TRUE),
    .groups = "drop"
  )

# =====
# STEP 5: Build Florida Master Data Frame
# =====
bea_aggregate <- bea_clean %>%
  select(Year, nominal_gsp, real_gsp, pce, government_gdp,
    construction_gdp, total_employment, gdp_deflator) %>%
  distinct()

florida_master <- oews_by_naics %>%
  left_join(
    naics_ces_bea_crosswalk %>% select(naics_2digit, ces_series_id) %>% distinct(),
    by = "naics_2digit"
  ) %>%
  left_join(
    ces_clean %>% rename(ces_series_id = seriesID),
    by = join_by(Year, ces_series_id)
  ) %>%
  select(-ces_series_id) %>%
  left_join(qcew_by_naics, by = join_by(Year, naics_2digit)) %>%
  left_join(bea_by_naics, by = join_by(Year, naics_2digit)) %>%
  left_join(laus_clean, by = "Year") %>%
  left_join(bea_aggregate, by = "Year")

glimpse(florida_master)

message("Years in florida_master: ", paste(sort(unique(florida_master$Year)), collapse =
", "))
message("Total rows: ", nrow(florida_master))

# =====
# SAVE
# =====
saveRDS(florida_master, "data/florida_master.rds")
message("Saved: data/florida_master.rds")

```